

Erdős Problem #584

EP-584

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let G be a graph with n vertices and δn^2 edges. Are there subgraphs $H_1, H_2 \subseteq G$ such that

H_1 has $\gg \delta^3 n^2$ edges and every two edges in H_1 are contained in a cycle of length at most 6, and furthermore if two edges share a vertex they are on a cycle of length 4, and

H_2 has $\gg \delta^2 n^2$ edges and every two edges in H_2 are contained in a cycle of length at most 8.